

Parkinson's Disease & Other Movement Disorders

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Outlines

- **Parkinsonism**
- **Idiopathic Parkinson's disease.**
- **Parkinson Plus syndromes.**
 - **MSA**
 - **PSP**
 - **CBD**
 - **LBD**
 - **HD**
 - **Wilson's disease**
- **Essential Tremor**
- **Hemifacial spasm**
- **Video presentation of clinical patterns.**

Objectives

The student must be able to:

- Diagnose Idiopathic Parkinson's disease & differentiate it from other Parkinson plus syndromes.
- Diagnose the main movement disorder patterns through video teaching.
- Treat idiopathic Parkinson's disease & some of the other common movement disorder like essential tremor.
- Understand the mechanisms of action of anti-Parkinsonian medications.
- Recognize the main side effects of anti-Parkinsonian medications.

Movement disorders

- Movement disorders present with a wide range of symptoms.
- They may be genetic or acquired, & the most important is Parkinson's disease.
- Most movement disorders are categorized clinically, with few confirmatory investigations available other than for those with a known gene abnormality.

Parkinsonism

- **Parkinsonism** is a clinical syndrome characterised by:
 - **Bradykinesia (slowing of Movement)**
 - **Rigidity (increased tone through-out movement),**
 - **Tremor &**
 - **Loss of postural reflexes.**
- More than 80% of cases are due to Idiopathic Parkinson's disease (PD).
- It may be caused by:
 - drugs,
 - degenerative disease (e.g. Alzheimer's),
 - anoxic or other injuries, or
 - genetic conditions (e.g. Huntington's disease),

Causes of Parkinsonism

- **Idiopathic Parkinson's disease**
(at least 80% of parkinsonism)
- **Cerebrovascular disease**
- **Anoxic brain injury**
- **Drugs & toxins**
 - Antipsychotic (older & 'atypical')
 - Metoclopramide
 - Prochlorperazine
 - Tetrabenazine
 - Sodium valproate
 - Lithium
 - Manganese
 - MPTP
- **Other degenerative diseases**
 - Dementia with Lewy bodies
 - Progressive supranuclear palsy
 - Multiple system atrophy
 - Corticobasal degeneration
 - Alzheimer's disease
- **Genetic**
 - Huntington's disease
 - Fragile X tremor ataxia syndrome**
 - Dopa-Responsive Dystonia
 - Spinocerebellar Ataxias
(particularly SCA 3)
 - Wilson's disease

Idiopathic Parkinson's disease

- PD is a progressive & incurable condition, with a variable prognosis.
- Whilst motor symptoms are the usual presenting features, cognitive impairment, depression & anxiety become increasingly common as the disease progresses, & significantly reduce quality of life.
- PD has an annual incidence of 18/100 000 in the UK & a prevalence of 180/100 000.
- Average age of **onset is ~60 yrs; < 5% present under 40.**
- Single genes causing PD have been identified in a small proportion of cases.
- Having a first-degree relative with PD increases the risk by 2–3 times.

Clinical features: Motor features

- The presentation is usually asymmetrical, e.g. a **resting tremor** in an upper limb.
- Typical features of an established case include:

Bradykinesia: Slowness in initiating or repeating movements, impaired fine movements causing small handwriting (**micrographia**) & expressionless face (**hypomimia**).

Slow to start walking, reduced arm swing, small steps & tendency to run (**festination**).

Tremor: rest (**3-4 Hz**), less on action; starts in fingers/thumb & may affect arms, legs, feet, jaw & tongue.

Rigidity: Cogwheel type mostly ULs limbs; plastic (lead pipe) type mostly affects LLs.

General

- Expressionless face (hypomimia)
- Soft, rapid, indistinct speech (dysphonia)
- Flexed (stooped) posture
- Impaired postural reflexes

Gait

- Slow to start walking (failure of gait ignition)
- Rapid, short stride length, tendency to shorten (festination)
- Reduction of arm swing
- Impaired balance on turning

Tremor

Resting (3–4 Hz, moderate amplitude): most common

- Asymmetric, usually first in arm/hand ('pill rolling')
- May affect legs, jaw and chin but not head
- Intermittent, present at rest, often briefly abolished by movement of limb, exacerbated by walking

Postural (6–8 Hz, moderate amplitude)

- Present immediately on stretching out arms

Re-emergent tremor (3–4 Hz, moderate amplitude)

- Initially no tremor on stretching arms out, rest tremor re-emerges after a few seconds

Physical signs in Parkinson's disease

Rigidity

- Cogwheel type, mostly upper limbs (due to tremor superimposed on rigidity)
- Lead pipe type

Akinesia (fundamental feature)

- Slowness of movement
- Fatiguing and decrease in size of repetitive movements

Normal findings (if abnormal, consider other causes)

- Power, deep tendon reflexes, plantar responses
- Eye movements
- Sensory and cerebellar examination

IPD Video



Parkinson's disease symptoms[via torchbrowser.com].mp4



Signs In Parkinsonism[via torchbrowser.com] (1).mp4



Non-motor symptom

Non-motor symptoms may precede typical motor symptoms by many years & include:

- Cognitive impairment: develops in 30-80%, of patients as the disease progresses.
- Depression.
- Anxiety.
- Apathy
- Hallucinoses / psychosis
- Sleep disturbance & hypersomnolence
- Fatigue
- Pain
- Sphincter disturbance & constipation
- Sexual problems (erectile failure, loss of libido or hypersexuality)
- Drooling &
- Weight loss.

BOF Question:

The motor symptom of Parkinson's disease include:

- A. Apathy.
- B. Fatigue.
- C. Hallucination.
- D. Festination.

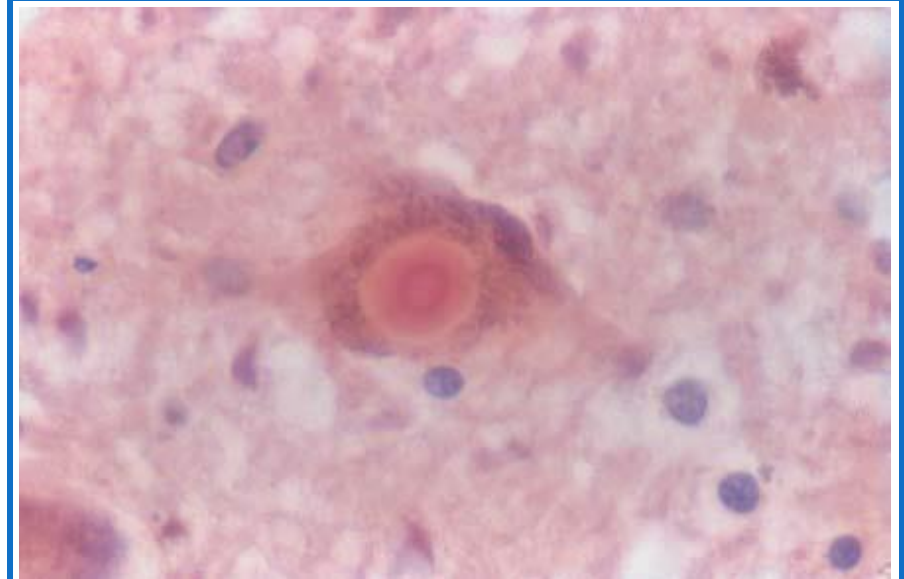
BOF Answer:

The motor symptom of Parkinson's disease include:

- A. Apathy.
- B. Fatigue.
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- D. Festination.**

Pathophysiology

- The pathological hallmarks of PD are depletion of the dopaminergic neurons in the **substantia nigra** & the presence of **α -synuclein** & other protein **inclusions** in nigral cells (**Lewy bodies**).
- It is thought that environmental or genetic factors alter the α -synuclein protein, rendering it toxic.
- PD is recognized as a **synucleinopathy**.*



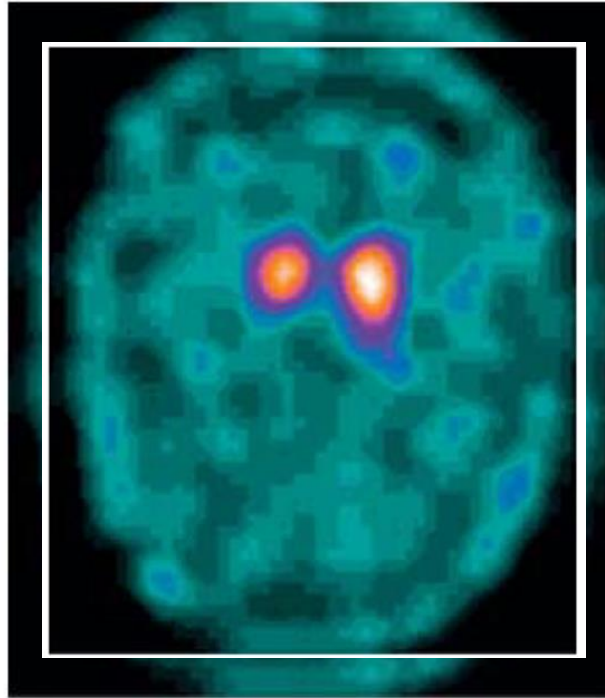
Substantia nigra of a patient with Parkinson's disease showing classical **Lewy body**

Investigations

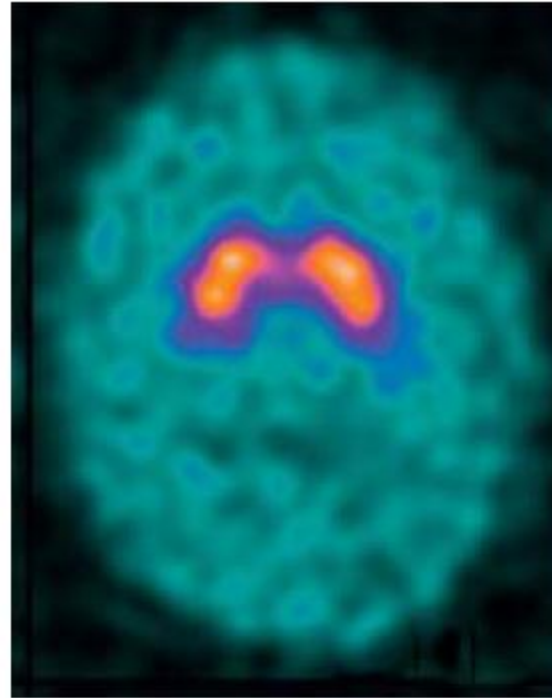
- **Diagnosis is clinical.**
- Neuroimaging (**CT or MRI**) may be needed if any features suggest pyramidal, cerebellar or autonomic involvement, or if the diagnosis is in doubt, but is usually normal for age.
- Functional dopaminergic imaging (**SPECT or PET**) is abnormal, even in the early stages but not specific for PD.
- Patients **< 50 yrs** should be tested for **Wilson's disease & Huntington's disease**.

SPECT in PD

A



B



A: Single photon emission computed tomography (SPECT) in Parkinson's disease showing reduced dopamine activity in the basal ganglia.

B: Normal.

MANAGEMENT of PD:

Pharmacologic Therapy in PD

1. Levodopa (L-dopa):

- Levodopa remains the **mainstay** of Rx, with a peripheral-acting dopa-decarboxylase inhibitor.
- Therapy should be started when symptoms are **impacting on daily life**.
- Decarboxylase inhibitors (**carbidopa & benserazide**) reduce peripheral side effects & are available as combination preparations with levodopa (**Sinemet & Madopar**).
- Levodopa is particularly effective at improving **bradykinesia & rigidity**.

Side-effects of Levodopa

- Postural hypotension.
- Nausea & vomiting.
- Involuntary movements (orofacial dyskinesias, limb & axial dystonias).
- Occasionally, depression, hallucinations & delusions.

Late deterioration despite levodopa occurs after 3–5 yrs in up to **50%** of patients. The simplest form is **end-of-dose** deterioration; this may be improved by using smaller, more frequent doses or a slow release preparation.

More complex fluctuations present as periods of alternating dyskinesia & agitation with severe parkinsonism (the **‘on–off’ phenomenon**).

Mechanisms of Drug Action in Parkinson's Disease

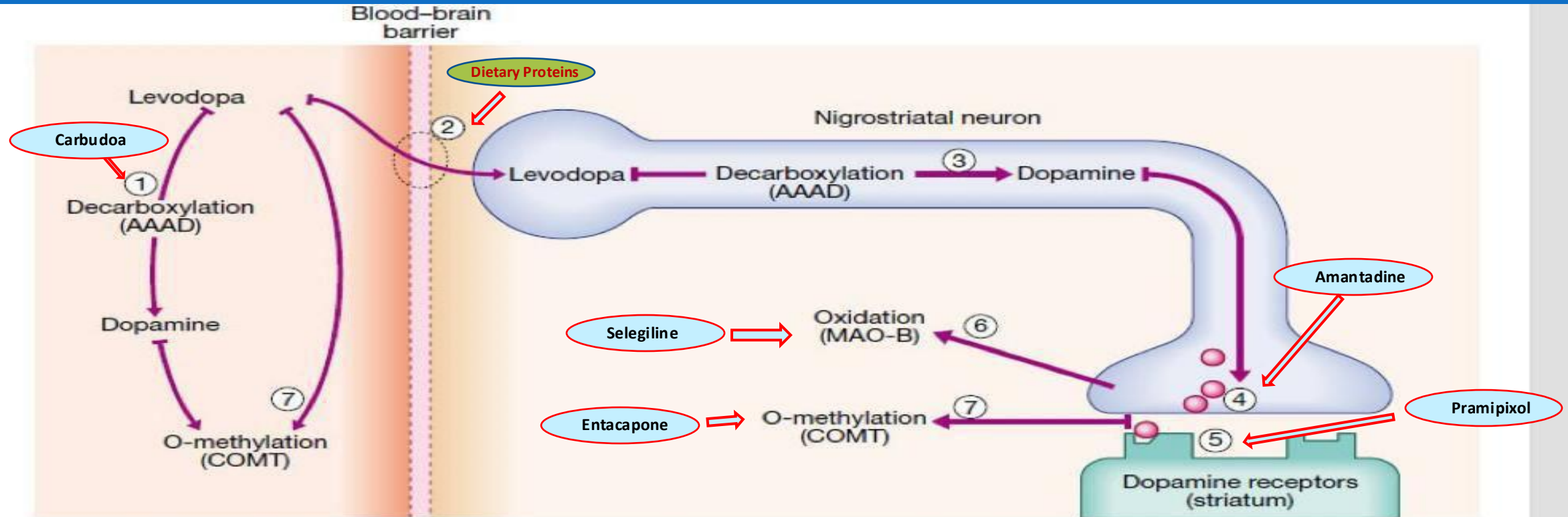


Fig. 26.42 Mechanisms of drug action in Parkinson's disease. (1) Decarboxylase inhibitors (carbidopa and benserazide) decrease side-effects by reducing peripheral conversion of levodopa to dopamine by aromatic amino acid decarboxylase (AAAD). (2) Active transport of levodopa into the brain may be inhibited by competition from dietary amino acids after a high-protein meal. (3) In the nigrostriatal neurons, levodopa is converted into dopamine. (4) Amantadine enhances the release of dopamine at the nerve terminal. (5) Dopamine agonists act directly on striatal receptors. (6) The monoamine oxidase type B (MAO-B) inhibitor selegiline increases the availability of neuronal dopamine by reducing its metabolism outside the neuron. (7) The catechol-O-methyl-transferase (COMT) inhibitor entacapone prolongs the availability of dopamine by inhibiting the metabolism of dopamine and levodopa outside the neuron.

Non-Dopaminergic Therapy for PD:

- 1) Dopamine receptor agonists (**Pramipixol & Rotigoten**).
- 2) MAOI-B (Monoamine oxidase type B) inhibitors (**selegiline & rasagiline**).
- 3) COMT (Catechol-O-methyl-transferase) inhibitors (**Entacapone & Tolcapone**).
- 4) Anti-glutaminergic therapy (**Amantadine**).
- 5) Anticholinergic agents: (Trihexyphenidyl [**Artane**] & Procyclidine [**Kemadrin**]).

Pharmaco - Therapy in PD

No	Agent	Mode of action	Side effects
1.	Levodopa (L-dopa):	Dopamine substitution	● Postural hypotension. ● Nausea & vomiting. ● Involuntary movements (orofacial dyskinesias, limb & axial dystonias). ● Occasionally, depression, hallucinations & delusions.
2.	Carbidopa, Benserazide	DOPA Decarboxylase inhibitors NB: Usually these 2 classes (1 & 2) are combined together to minimize side effects	
3.	Oral options include: ▪ bromocriptine, ▪ pergolide, ▪ cabergoline, ▪ lisuride, ▪ ropinirole & ▪ pramipexole Parenteral option include: ▪ Rotigotene transderal ▪ Apomorphine	Dopamine receptor agonists	● Nausea ● Vomiting ● Confusion ● Hallucinations
4.	selegiline & rasagiline	MAOI-B (Monoamine oxidase type B) inhibitors	
6.	Entacapone Tolcapone	COMT (Catechol-O-methyl-transferase) inhibitors:	
7.	Amantadine	Anti-glutaminergic	● Livedo reticularis ● Peripheral edema ● Confusion & ● seizures
8.	Trihexyphenidyl (Artane) Procyclidine (Kemadrin)	Anticholinergic	● dry mouth ● blurred vision ● constipation ● urinary retention ● confusion & hallucinosis

Management: non-pharmacological

Surgery:

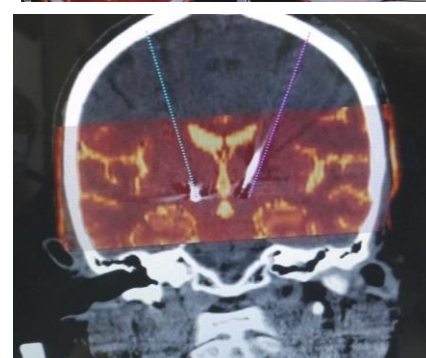
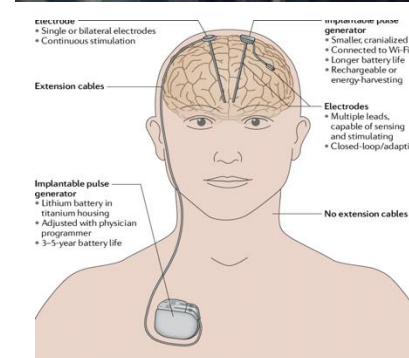
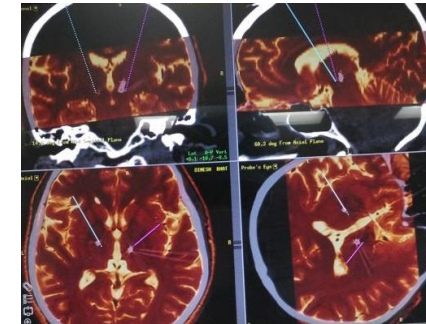
- Stereotactic surgery most commonly involves **deep brain stimulation (DBS)**, rather than the destructive approach of previous eras.
- Various targets have been identified, including:
 - **Subthalamic nucleus (STN)**,
 - Globus pallidus &
 - Thalamus (only effective for tremor)

DBS is usually reserved for patients with:

- **Medically refractory tremor or**
- **Motor fluctuations.**

Physiotherapy: reduces rigidity & corrects abnormal posture.

Speech therapy: may help dysarthria & dysphonia.



BOF

The unlikely statement regarding levodopa therapy is:

- A. Levodopa remains the mainstay of treatment of Parkinson's disease.
- B. Therapy should be started when symptoms are impacting on daily life.
- C. Levodopa is particularly effective at improving bradykinesia & rigidity.
- D. Livedo reticularis & seizures are common side effects.

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DBS video



Amazing DBS Before & After - 225-769-2200 - Baton Rouge Parkinson's Specialists[via torchbrowser.com].mp4



Other Parkinsonian Syndromes

- CVA & drug-induced parkinsonism are the most common alternative causes of parkinsonism.
- There are several degenerative conditions that cause parkinsonism, including:
 - **multiple systems atrophy,**
 - **progressive supranuclear palsy &**
 - **corticobasal ganglionic degeneration.**
- They typically have a more rapid progression than PD & tend to be resistant levodopa Rx .

Multiple systems atrophy (MSA)

- **MSA** is characterised by parkinsonism, autonomic failure & cerebellar symptoms, with either parkinsonism (MSA-P) or cerebellar features (MSA-C) predominating.
- Diagnosis is assisted by autonomic function tests.
- Falls are more common than in idiopathic Parkinson's disease, & life expectancy is reduced.
- Cognition is usually unaffected
- Is an α -synucleinopathy
- **Rx** is symptomatic & the prognosis is worse than PD

Progressive supranuclear palsy (PSP)

Clinical features of PSP include:

- **Symmetrical parkinsonism**
- Rigidity in extension rather than flexion
- **Supranuclear** paralysis of eye movements.
- Other features are pyramidal signs & cognitive impairment.
- PSP is associated with abnormal accumulation of **tau** (τ) proteins, therefore it is a **tauopathy** rather than synucleinopathy.
- There is **no treatment**, & the parkinsonism usually does not respond to L-Dopa.

Corticobasal degeneration (CBD)

- **CBD** is less common than MSA or PSP, & the clinical manifestations include:
 - **parkinsonism,**
 - **dystonia,**
 - **myoclonus &**
 - **‘alien limb’ phenomenon, whereby a limb (usually upper) moves about or interferes with the other limb without apparent conscious control.**
- Cortical symptoms, including dementia & especially apraxia, are common & may be the only features in some cases.
- CBD is a **tauopathy** widespread deposition throughout the brain.
- Similar survival rates to MSA & PSP.

Causes of Tauopathy vs. α -synucleinopathy

tauopathy	α-synuclein
Alzheimer dis	Idiopathic PD
PSP	MSA
CBD	LBD

Alien limb phenomenon in CBD



Cycling for Freezing Gait in Parkinson's Disease[via torchbrowser.com].mp4

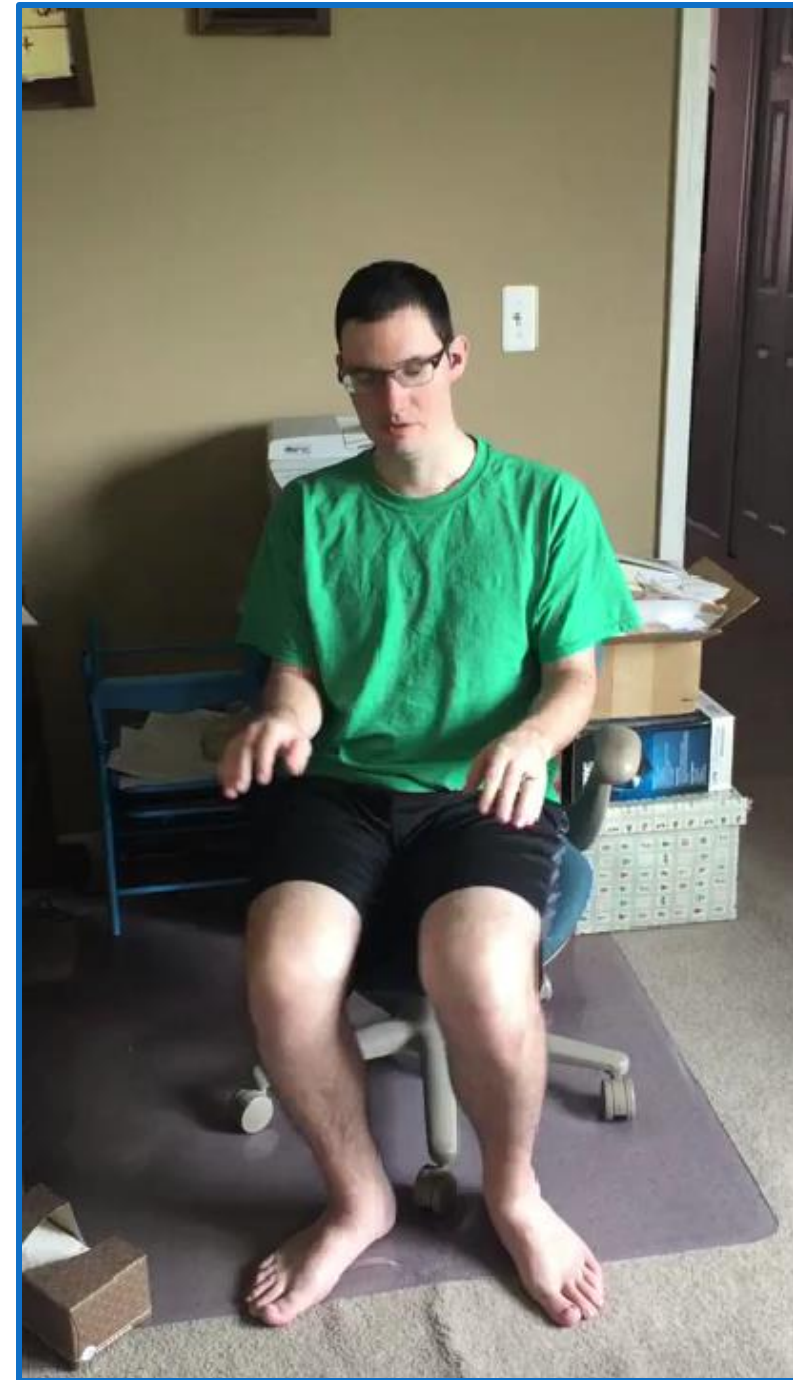
Wilson's disease

- An autosomal recessive dis. due to mutation in the *ATP7B* gene, causing defective copper metabolism.
- It is a treatable cause of various movement disorders, including:
 - ✓ tremor,
 - ✓ dystonia,
 - ✓ Parkinsonism,
 - ✓ ataxia;
 - ✓ psychiatric symptoms may also occur.
- Wilson's disease must always be excluded in patients **under 50 year** with any movement disorder.

Wilson's disease Video



Wilson Disease Bad Times[via torchbrowser.com].mp4



Huntington's disease (HD)

- HD is an autosomal **dominant** inherited disorder, usually presenting in adult life.
- It is due to expansion of a **trinucleotide** repeat on **chromosome 4** & frequently demonstrates anticipation (younger age of onset in subsequent generations).

Clinical Features:

- Gradually progressive chorea & behavioral disturbance are the earliest symptoms.
- There is cognitive impairment, which later becomes frank dementia.
- Seizures may occur late in the disease.
- The **Westphal variant** manifest as hypokinetic-rigid (parkinsonism) syndrome often with juvenile onset.

Investigations of HD:

- The diagnosis is confirmed by **genetic** testing; pre-symptomatic testing for other family members is available but must be preceded by appropriate counselling.
- Brain imaging may show **caudate atrophy** but is not reliable.

Management of HD:

- Chorea may respond to **risperidone** or **tetrabenazine**.
- Depression is common & may be helped by **antidepressant** medication.
- Long-term psychological support & eventually institutional care are often needed.

Essential tremor

- This has a prevalence of about 300/100 000 & may display a **dominant** pattern of inheritance, although no genes have thus far been identified.
- It may present at any age with a bilateral arm tremor (**8–10 Hz**), rarely at rest but typical with movement.
- The head & voice may be involved.
- The tremor improves in about 50% of patients with small amounts of alcohol.
- No specific tests & essential tremor should be distinguished from other tremors, including dystonic tremor.

Rx:

- Beta-blockers & primidone, topiramate or gabapentine are sometimes helpful, &
- DBS of the thalamus is an effective treatment for severe cases.

Hemifacial spasm

- This usually presents after middle age.
- Intermittent twitching around one eye, spreading ipsilaterally to other facial muscles.
- The spasms are exacerbated by talking, eating & stress.
- Hemifacial spasm is usually **idiopathic**, similar to trigeminal neuralgia.
- It has been suggested that it may be due to an aberrant arterial loop irritating the 7th nerve just outside the pons.
- It may, however, be **symptomatic** & secondary to structural lesions.

Rx:

- Drug treatment is not effective (Baclofen , Carbamazepine) but injections of **botulinum** toxin into affected muscles help, although these usually have to be repeated every 3 months or so.
- In refractory cases, microvascular decompression may be considered.

BOF Question

The unlikely statement regarding hemifacial spasm is:

- A. Hemifacial spasm usually presents after middle age.
- B. Injection of botulinum toxin is not a helpful treatment.
- C. The spasms are exacerbated by talking, eating & stress.
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Video of hemifacial spasm

60 second video for surgeons - HFS-hemifacial spasm(via touchbrowsers.com)mp4



Conclusions

- PD is degenerative chronic treatable condition. .
- PD can be differentiated clinical at time of presentation from other parkinsonian syndrome.
- Levodopa is the corner stone for treatment of PD.
- Other parkinsonian syndromes are unresponsive to levodopa therapy.
- Wilson disease may present with any type of movement disorder.
- Essential tremor is AD & responsive to medical Rx.
- HD is more aggressive disorder with eventual residential institutionalization.

Conclusions

Thanks a lot for your attention